Department of Electrical Engineering and Department of Computer Science Engineering
GURU NANAK DEV ENGINEERING COLLEGE
LUDHIANA • PUNJAB • EST. 1956 • NAAC 'A' GRADE

ONE MONTH INDUSTRIAL TRAINING

Linux, Programming & MATLAB

A practical, hands-on training in open-source tools & engineering computation for tomorrow's engineers.

4 / 6 WEEKS

DURATION	MODE	FEE
4 / 6 Weeks	**Offline**	**Rs. 4000 / 5500**
Tentative Jun–Jul 2026	Hands-on Lab Sessions	Inclusive of GST

ABOUT THE COLLEGE

Guru Nanak Dev Engineering College (GNDEC), Ludhiana, established in 1956, is one of the oldest and premier Engineering Institutes of India. Spread across 88 acres along the Gill Road, the foundation stone was laid by Honorable Dr. Rajendra Prasad Ji, then President of India. GNDEC is an autonomous college under UGC Act 1956 and is NAAC-accredited with 'A' grade. The institution offers thirteen UG, ten PG programmes and is an IKGPTU autonomous college Centre for Ph.D.

OBJECTIVES

- Build a practical foundation in the Linux ecosystem and command-line workflows.
- Develop scripting skills in Bash and Python for engineering automation.
- Use Git and GitHub for version-controlled collaborative work.
- Apply MATLAB for matrix computation, plotting and problem solving.
- Use MATLAB / Simulink for signal processing, control systems and circuit simulation.
- Execute a Capstone Project (6-week track) and document it professionally.

WHO CAN APPLY

- B.Tech students of EE / ECE / CSE / IT / ME / Civil branches.
- Students of 2nd and 3rd year of B.Tech are eligible.
- BCA and MCA students are also welcome to apply.
- Students from other Engineering / Polytechnic colleges may also enroll.

ORGANIZED BY
Guru Nanak Dev Engineering College, Ludhiana
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CURRICULUM

What You'll Learn

Six weeks. Six themes. From terminal to capstone.

4-WEEK CORE TRACK

01 LINUX FUNDAMENTALS

Ubuntu installation, filesystem, shell commands, permissions, processes, package management, pipes and redirection.

02 PROGRAMMING & GIT

Bash scripting essentials, Python with NumPy & Matplotlib, version control with Git and GitHub workflow.

03 MATLAB FUNDAMENTALS

Vectors, matrices, control flow, scripts & functions, 2D/3D plotting and file I/O for engineering math.

04 MATLAB APPLICATIONS

Symbolic math, FFT & filtering, transfer functions, Bode plots, intro to Simulink with control modeling.

6-WEEK EXTENSION TRACK

05 ADVANCED MATLAB & SIMULINK

Structures, performance tuning, Simulink subsystems, power electronics, modulation and machine modeling.

06 CAPSTONE PROJECT

Team-based MATLAB/Simulink mini-project — design, simulation, final report and live demo presentation.

TRAINING MENTORS

Meet Your Mentors

K **Pf. Karanbir Singh**
EE • +91 88727 13231

S **Pf. Sukhpal Singh**
EE • +91 97794 22219

K **Pf. Kuldeep Singh Kooner**
EE • +91 97793 48866

S **Dr. Satinderpal Singh**
CSE • +91 82890 33132

For queries & registration, contact:
karanbir@gndec.ac.in

HOW TO REGISTER

Reserve Your Seat

1. Fill the registration form.
2. Email scanned form to coordinator.
3. Pay the fee (4-wk / 6-wk).
4. Submit hard copy on Day 1.

FEES (INCL. GST)
Rs. 4000 / 4-Week • Rs. 5500 / 6-Week

REGISTRATION

Registration Form

Please fill all fields in BLOCK LETTERS using a black or blue pen.

Name (in block letters): ____________________

University Roll Number: ____________________ **College Roll Number:** ____________________

Department: ____________________ **Year of Study:** ____________________

Institution / College: ____________________

Mailing Address: ____________________

____

Age: ____________ **Gender (M/F/O):** ____________ **Date of Birth:** ____________

Mobile No.: ____________________ **Email:** ____________________

TRACK & FEE (tick one)

☐ 4-Week Track **Rs. 4000/-** (Including GST)

☐ 6-Week Track **Rs. 5500/-** (Including GST)

Receipt No. (Cash Payment): ____________________ **Date:** ____________________

Signature of Applicant	Departmental T&P Coordinator	Date & Place

IMPORTANT NOTE

Participants must email the scanned copy of the filled and duly signed registration form, along with the fee receipt, to karanbir@gndec.ac.in on or before the announced deadline. The hard copy of the registration form and fee receipt must be submitted on the first day of the program at the registration desk.